Homework Assignment 1

Q1-Q3, You can find solutions from PPT Lec01 directly.

Question 1 (10 points):

Describe the relative merit of procedural and non-procedural languages.

Non-procedural languages greatly simplify the specification of queries (at least, the types of queries they are designed to handle). They free the user from having to worry about how the query is to be evaluated; not only does this reduce programming effort, but in fact in most situations the query optimizer can do a much better task of choosing the best way to evaluate a query than a programmer working by trial and error. On the other hand, procedural languages are far more powerful in terms of what computations they can perform. Some tasks can either not be done using nonprocedural languages or are very hard to express using nonprocedural languages.

Question 2 (10 points):

Explain the difference between physical and logical schemas.

Physical schemas describe the database design at the physical level which represents how the data are actually stored on storage media. Logical schemas describe the database design at the logical level which specifies what data are stored and what relationships exist among those data.

Question 3 (10 points):

Describe the differences in meaning between the term “relation” and “relation schema”.

A relation is an unordered set of tuples that share the same type. A relation schema is the type definition (i.e., a set of attributes with defined domains) of a relation.

Question 4 (10 points)

Describe 3 tables that might be used to store information in a social networking system such as Facebook. Please name your tables and describe their relation schemas.

• A users table containing records about users, with attributes such as account name, real name, age, gender, location, and other profile information.

• A content table containing user provided content, such as text and images, associated with the user who uploaded the content.

• A friends table recording for each user which other users are connected to that user.

Question 5 (10 points)

Consider the advisor relation shown in the figure below, with s_id as the primary key of advisor. Suppose a student can have more than one advisor. Then, would s_id still be a primary key of the advisor relation? If not, what should the primary key of advisor be?

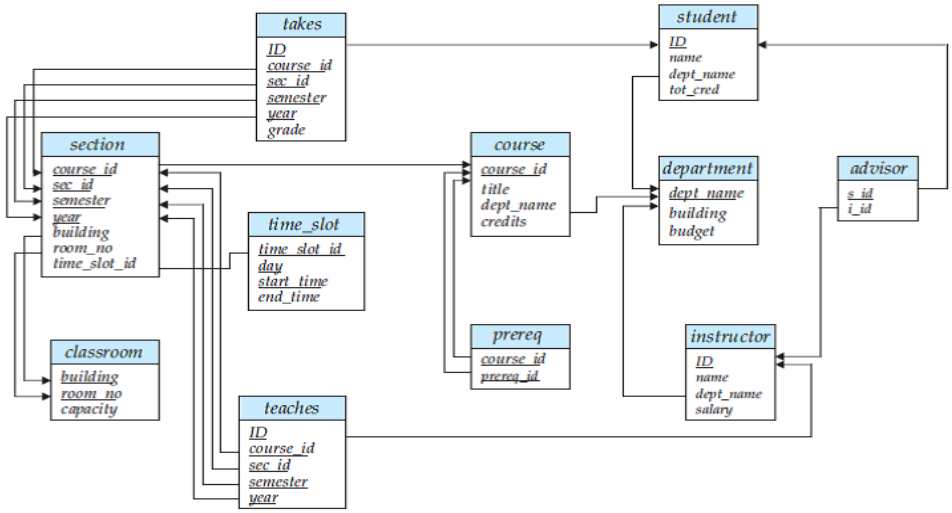

Figure 2.8 Schema diagram for the university database.

Question 6 (30 points):

Consider the following relational database:

student(student_id, name)

course(course_id, classroom)

take(student_id, course_id)

(1) Identify an appropriate primary key for each relation schema.

student(student id, name)

course(course id, classroom)

take(student id, course id)

The primary key for student is student id since each student must have a unique ID, but not a unique name. The primary key for course is course id since each course must have a unique ID, but not a unique

classroom number. The primary key for take is {student id, course id} since every combination of a student and a course must be unique, but a student can take multiple courses and a course can be taken by multiple students.

(2) Given your choice of primary keys, identify all foreign keys

take: student id referencing student and course id referencing course. Answers that do not specify the referenced relations (e.g., the student relation when student id is considered as a foreign key in the take relation) are incomplete, so will lose points.

Student and Course relations have no foreign key.

(3) Using the above relation schemas, provide an example of a superkey which is not a candidate key.
{student id, name} is a superkey for the student relation (since id+name can identify each unique record), but is not a candidate key because student id is a superkey (i.e., {student id, name} is not a minimal superkey).

Question 7 (30 points):

Consider the database in Problem 6. Give an expression in the **relational algebra** for each of the following queries:

(a) (10 points) Find the IDs of all students whose name is John Smith.

$\Pi_{student_id}(\sigma_{name=\text{“}JohnSmith''}(student))$

(b) (10 points) Find the IDs of the courses that John Smith takes.

$\Pi_{course_id}(\sigma_{name=\text{“}JohnSmith''}(student) \bowtie take)$

(c) (10 points) Find the classrooms used for the courses that John Smith takes.

$\Pi_{classroom}(\sigma_{name=\text{“}JohnSmith''}(student) \bowtie take \bowtie course)$